ASSIGNMENT 4
Apply Naïve Bayes Classification Algorithm

1. Aim

To implement the Gaussian Naïve Bayes classification algorithm and evaluate its performance using the Iris dataset.

2. Objectives

- To understand supervised machine learning.
- To understand probability-based classification.
- To implement the Gaussian Naïve Bayes algorithm.
- To predict the class labels of test data.
- To evaluate classification performance using accuracy and other metrics.

3. Theory

Classification is a supervised machine-learning task in which a model learns from labelled examples and predicts the category of new observations.

Naïve Bayes is a classification algorithm based on Bayes' theorem. It calculates the probability of a class given the observed features and predicts the class with the highest estimated probability.

The algorithm is called naïve because it assumes that features are conditionally independent given the class.

Bayes' theorem is expressed as:

$$P(C \mid X) = \frac{P(X \mid C)P(C)}{P(X)}$$

Where:

- P(C | X) is the probability of class C given the observed features.
- P(X | C) is the likelihood of observing the features for that class.
- P(C) is the prior probability of the class.
- P(X) is the overall probability of the observed features.

Gaussian Naïve Bayes is suitable for numerical features when their distributions within each class are modelled using Gaussian distributions.

The Iris dataset contains measurements of flowers and labels representing three species. The dataset is divided into training and testing sets. The training set is used to learn the model, while the testing set is used to evaluate its predictions.

Accuracy measures the proportion of test examples classified correctly. Precision, recall, and F1-score provide additional information about classification performance.

4. Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate the features and target labels.
4. Divide the dataset into training and testing sets.
5. Create a Gaussian Naïve Bayes model.
6. Train the model.
7. Calculate accuracy and display the classification report.

5. Explanation of Important Code

- load_iris() loads the flower dataset.
- X contains the flower measurements, and y contains the species labels.
- train_test_split() divides the data into training and testing subsets.
- GaussianNB() creates the Naïve Bayes classifier.
- fit() trains the model using the training data.
- predict() predicts the class of each test observation.
- accuracy_score() calculates the proportion of correct predictions.
- classification_report() displays precision, recall, F1-score, and support.

6. Applications

1. Email spam detection.
2. Document and news classification.
3. Sentiment analysis.
4. Basic medical classification research.
5. Text categorisation systems.

7. Advantages

1. Simple to implement.
2. Usually fast to train.

3. Can work well with relatively small datasets.
4. Provides a probability-based approach to classification.

8. Limitations

1. The feature-independence assumption may not hold in real datasets.
2. Performance depends on the distribution of the input data.
3. Probability estimates may not always be well calibrated.
4. The Gaussian assumption may not suit every numerical dataset.

9. Conclusion

The Gaussian Naïve Bayes algorithm was implemented to classify Iris flowers into three species. The model's performance was evaluated using accuracy, precision, recall, and F1-score. The experiment demonstrated how probability-based classification can be applied to labelled data.